

Module 1: Lexical Analysis

Language Processors, Buffering, Thompson NFA & Direct DFA

Module I: Lexical Analysis — Topics Covered

1. Language Processing Systems & Cousins of Compiler
2. 6 Phases of Compiler with Running Example
3. Token, Pattern, Lexeme & Token Attributes
4. Input Buffering (Buffer Pairs & Sentinels)
5. Thompson's Construction (RE → NFA)
6. Subset Construction (NFA → DFA)
7. Hopcroft's DFA State Minimization
8. Direct DFA Construction (Syntax Tree Method)

1. Language Processors & 6 Phases of Compiler

A **compiler** translates high-level source code into target machine language while reporting errors.

Tool	Input → Output	Core Responsibilities
Preprocessor	<code>.c</code> → <code>.i</code> (Pure Source)	Macro expansion (<code>#define</code>), header inclusion (<code>#include</code>), stripping comments.
Compiler	<code>.i</code> → <code>.s</code> (Assembly)	Front-end analysis & back-end code synthesis.
Assembler	<code>.s</code> → <code>.o</code> (Relocatable Object)	Translates mnemonics to machine code and symbol offsets.
Linker	<code>.o</code> + Libs → <code>.exe</code>	Resolves external symbol references across modules.
Loader	Binary → RAM	Allocates memory segments, sets up stack/heap, jumps to <code>main</code> .

2. Input Buffering & Sentinels

Using **Buffer Pairs** ($2 \times N$ characters) with **Sentinel** ('EOF') characters placed at buffer ends eliminates one conditional check per character, reducing inner scanning loop overhead by 50%.

3. Direct DFA Construction (Syntax Tree Method)

Constructs DFA directly from augmented regular expression $(r)\#$:

- `nullable(n)` : True iff subtree generates ϵ .
- `firstpos(n)` : Set of positions matching the first character of subtree strings.
- `lastpos(n)` : Set of positions matching the last character of subtree strings.
- `followpos(i)` : Positions immediately following position i .
 - For cat-node $c_1 \cdot c_2$: add `firstpos( $c_2$ )` to `followpos( $i$ )` for all $i \in \text{lastpos}(c_1)$.
 - For star-node c_1^* : add `firstpos( $c_1$ )` to `followpos( $i$ )` for all $i \in \text{lastpos}(c_1)$.

 BIT Mesra Exam Solved Problem: $(a|b)^*abb\#$

Leaves: $a(1), b(2), a(3), b(4), b(5), \#(6)$.

$\text{followpos}(1) = \{1, 2, 3\}, \text{followpos}(2) = \{1, 2, 3\}, \text{followpos}(3) = \{4\}, \text{followpos}(4) = \{5\}, \text{followpos}(5) = \{6\}, \text{followpos}(6) = \emptyset$.

DFA States: $A = \{1, 2, 3\} \xrightarrow{a} B = \{1, 2, 3, 4\} \xrightarrow{b} C = \{1, 2, 3, 5\} \xrightarrow{b} D^* = \{1, 2, 3, 6\}$.